

DRO with Evidential Deep Learning

240806

Hyeongchan Ham

Contents

- Previous Work
 - Research comments
- Current Work
 - DRO with Evidential Deep Learning
- Future Work
 - 8. 8. (목) 14:00 KAIST EE 1차 발표

Research Comments

- DRO의 ambiguity set을 기존의 방식과 달리 evidential deep learning에서 나오는 Normal Inverse-Gamma (NIG) 분포를 사용하는 것을 제안하였음.
 - Comments: NIG 분포를 사용해야 하는 근거가 부족하다.

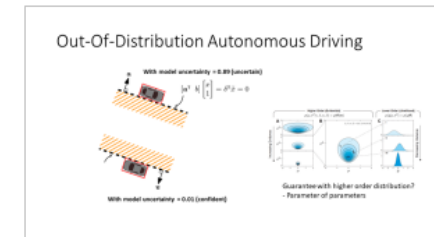
Distributionally Robust Optimization

- From the ambiguity set $\mathcal{D}$,
 - Normal Inverse-Gamma?

$$p(\underbrace{\mu, \sigma^2}_{\theta} | \underbrace{\gamma, v, \alpha, \beta}_m) = \frac{\beta^\alpha \sqrt{v}}{\Gamma(\alpha) \sqrt{2\pi\sigma^2}} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left\{-\frac{2\beta + v(\gamma - \mu)^2}{2\sigma^2}\right\}$$

- Optimize $\mathbf{x}$ based on the worst case distribution F

$$\underset{\mathbf{x} \in \mathcal{X}}{\text{minimize}} \left(\max_{F \in \mathcal{D}} \mathbb{E}_F[h(\mathbf{x}, \xi)] \right)$$



240724 presentation slide

Hakobyan, A., & Yang, L. (2021). Wasserstein distributionally robust motion control for collision avoidance using conditional value-at-risk. IEEE Transactions on Robotics, 38(2), 939-957.

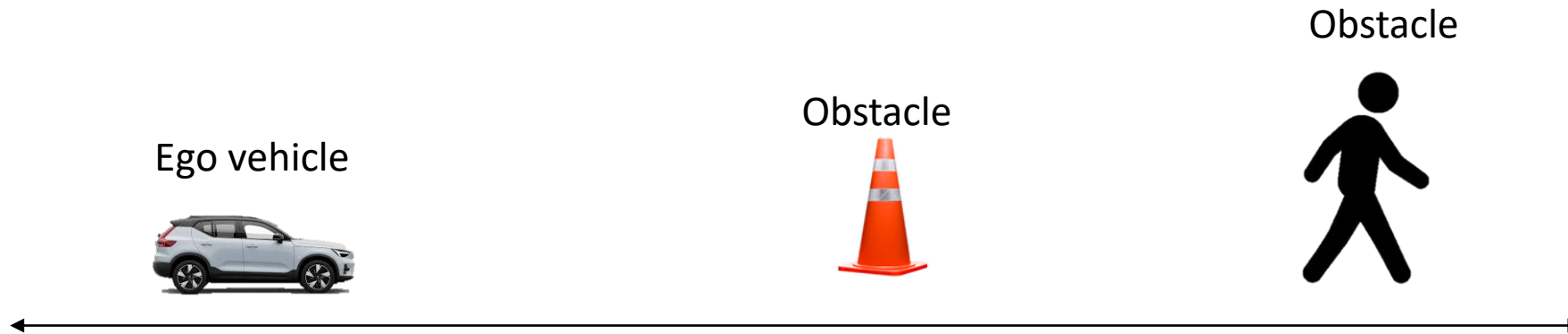
240731_lab meeting slide

DRO with Evidential Deep Learning

Hyeongchan Ham

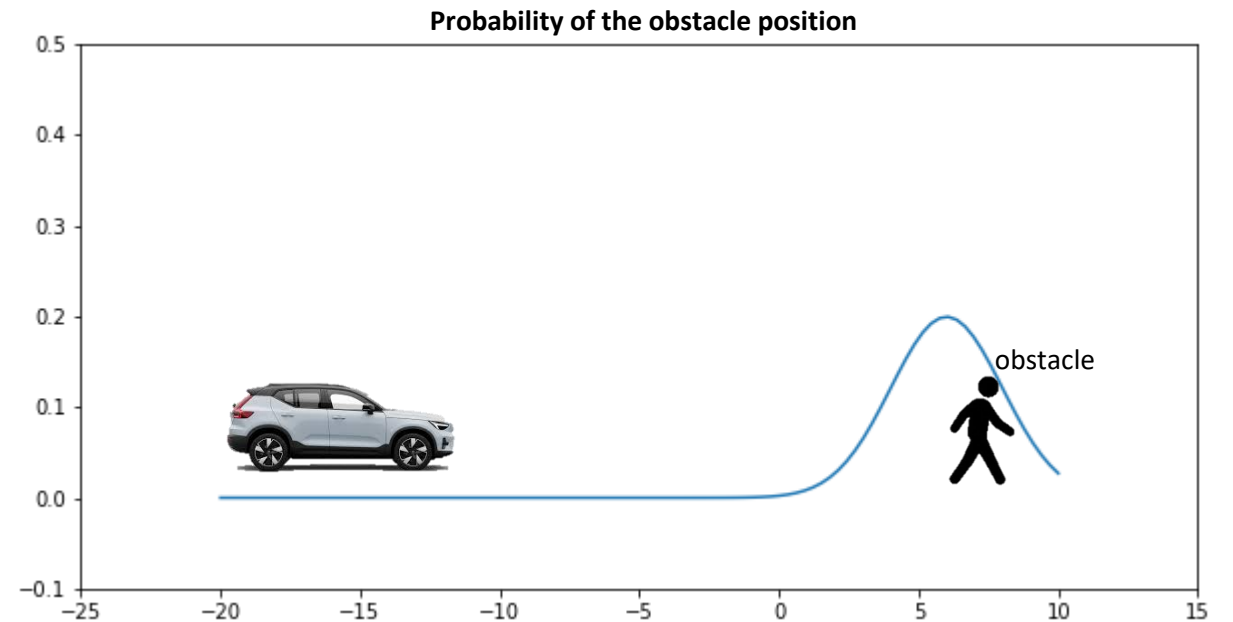
Introduction

- Safe operation of the mobile vehicles in uncertain environment is important.
- The agent (or the vehicle) perceives the surround obstacles and navigates the observed environment with no collision.



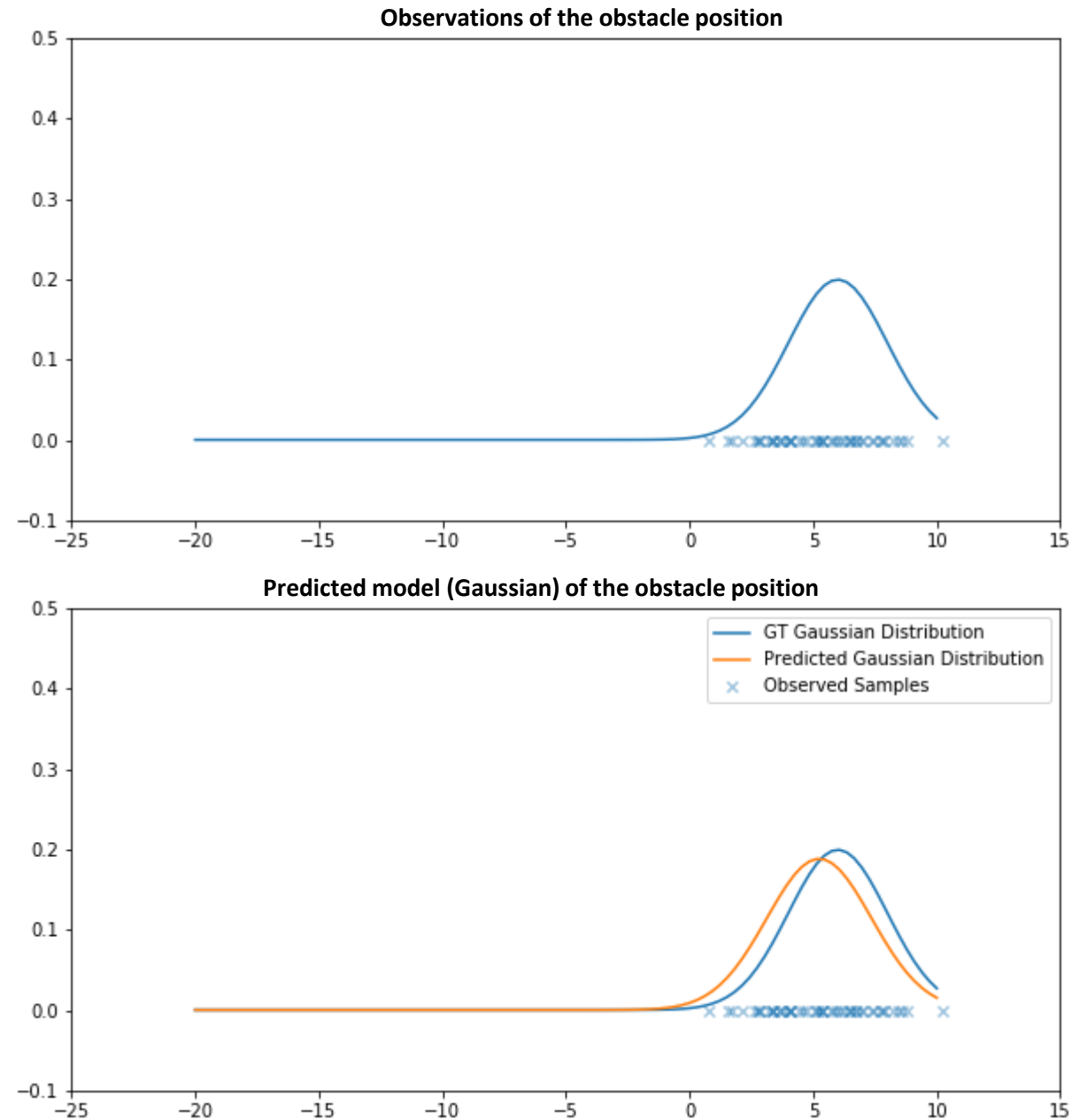
Introduction

- 1D example
 - We cannot perceive the exact state of the obstacles, so the probabilistic information is used.



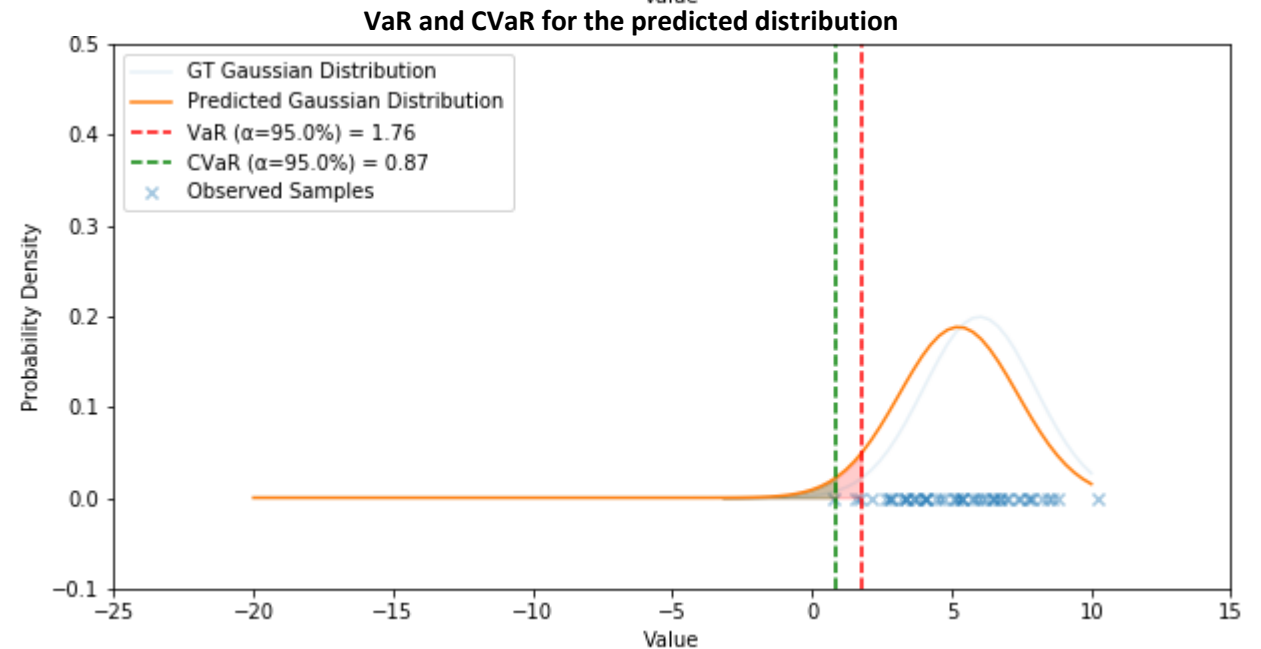
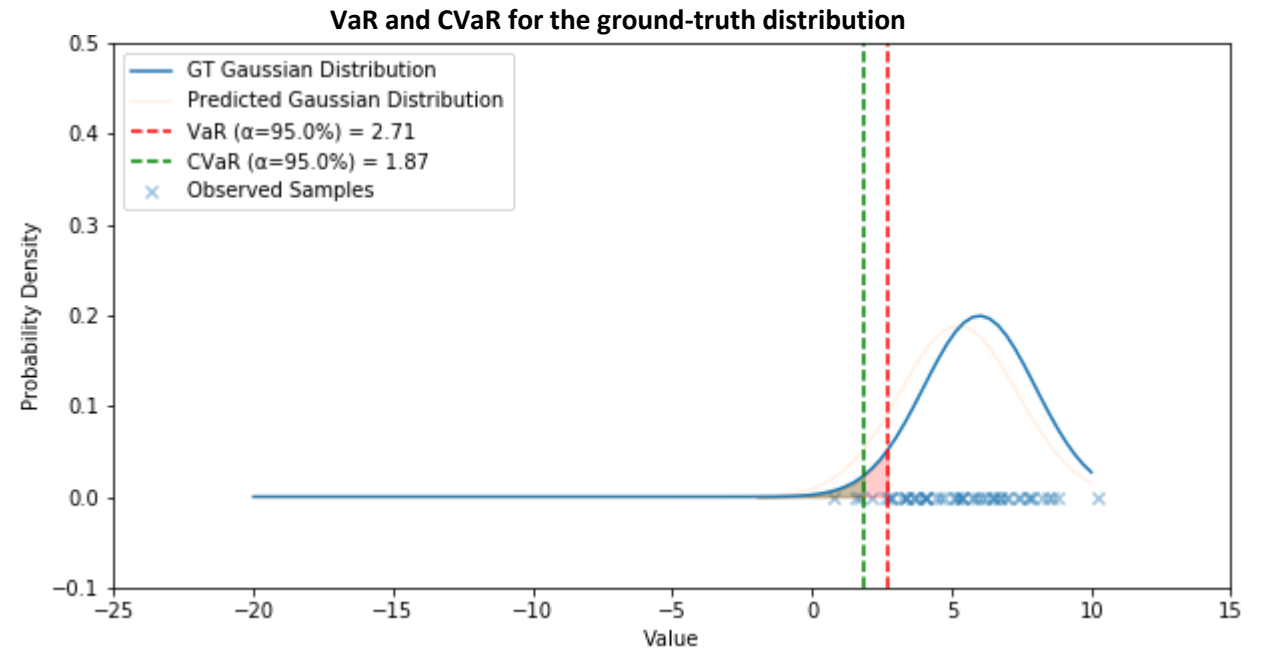
Introduction

- The perception module can only observe some samples of the obstacles, so we model the information about the probability distribution.



Introduction

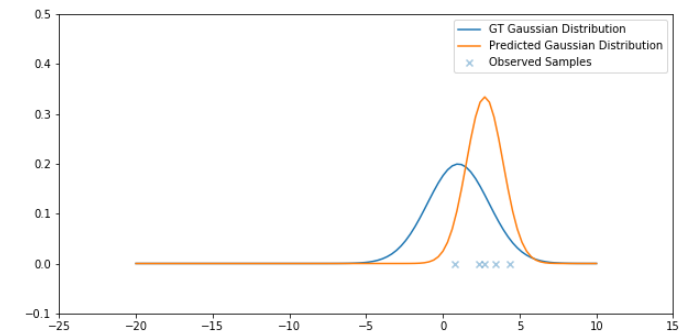
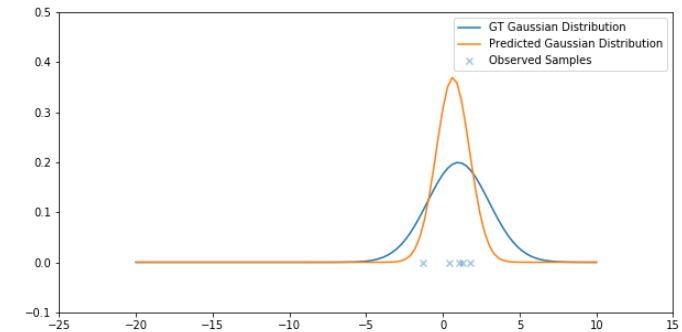
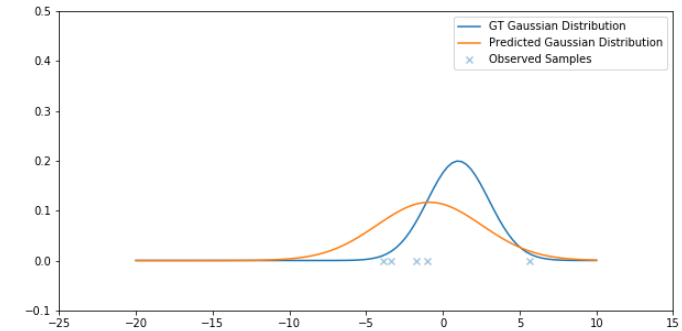
- We can use this models to make some constraints for the optimal control.
- For example, risk metrics such as VaR, CVaR can be used to construct the safe region from the obstacle.



Challenge

- However, it is challenging to obtain the correct model because they rely on the observed samples, which is the partial observation of the exact obstacles.
- **There exists model (epistemic) uncertainty.**

Predicted models (Gaussian) of the different obstacle position samples



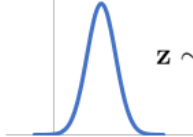
Solution

- Some previous works handle this problem with the *distributionally robust optimization (DRO)*, which chooses *the worst case* among *the ambiguity set*, and optimize based on this one.

Deterministic Optimization

$$\min_{\beta} h_{\beta}(\mathbf{z})$$

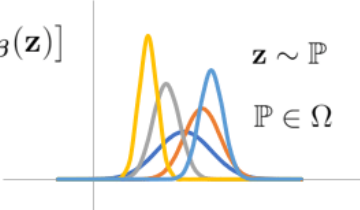

Stochastic Optimization

$$\inf_{\beta} \mathbb{E}^{\mathbb{P}}[h_{\beta}(\mathbf{z})]$$


Robust Optimization

$$\min_{\beta} \max_{\mathbf{z} \in \mathcal{Z}} h_{\beta}(\mathbf{z})$$


Distributionally Robust Optimization

$$\inf_{\beta} \sup_{\mathbb{P} \in \Omega} \mathbb{E}^{\mathbb{P}}[h_{\beta}(\mathbf{z})]$$


Comparison of robust optimization with distributionally robust optimization [1]

Related Works

- Moment based DRO
 - Ambiguity set is specified by the first, second moments. [2,3]
- Wasserstein based DRO
 - Ambiguity set is specified by the Wasserstein distance around the reference distribution. [4-6]
- 자료 조사가 더 필요합니다.

[2] Delage, E., & Ye, Y. (2010). Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Operations research*, 58(3), 595-612.

[3] Nie, J., Yang, L., Zhong, S., & Zhou, G. (2023). Distributionally robust optimization with moment ambiguity sets. *Journal of Scientific Computing*, 94(1), 12.

[4] Xu, S., Ruan, H., Zhang, W., Wang, Y., Zhu, L., & Ho, C. P. (2023). Wasserstein Distributionally Robust Chance Constrained Trajectory Optimization for Mobile Robots within Uncertain Safe Corridor. *arxiv*

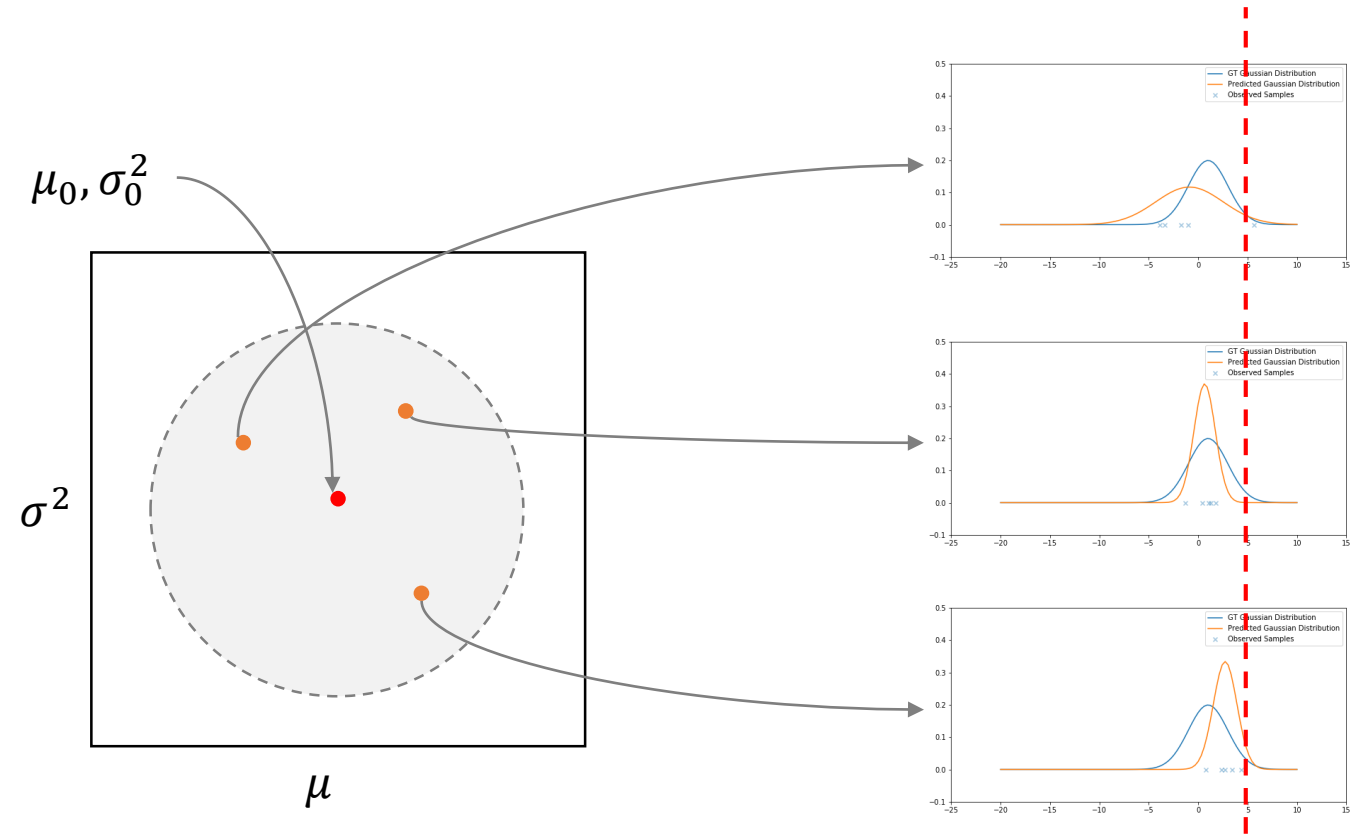
[5] Safaoui, S., & Summers, T. H. (2023). Distributionally Robust CVaR-Based Safety Filtering for Motion Planning in Uncertain Environments. *arxiv*

[6] Hakobyan, A., & Yang, I. (2023, May). Distributionally robust optimization with unscented transform for learning-based motion control in dynamic environments.

In 2023 IEEE International Conference on Robotics and Automation (ICRA) (pp. 3225-3232). IEEE.

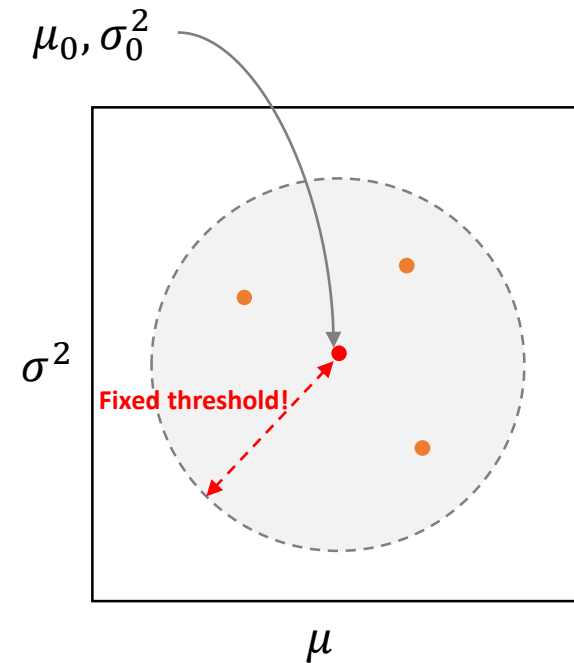
Related Works

- Wasserstein-based DRO
 - Ambiguity set is specified by the Wasserstein distance around the reference distribution (μ_0, σ_0^2) .



Limitations

- We can estimate the amount of the uncertainty, but current work does not consider this and treat all uncertainty as the same conservative level (same distance).



Motivation

- We can improve this by considering the measurement of the uncertainty estimation.
 - Large ambiguity set (more conservative) when the sample is unknown, less confident, and little knowledge.
 - Small ambiguity set (less conservative) when the sample is well-known, highly confident, and enough knowledge.

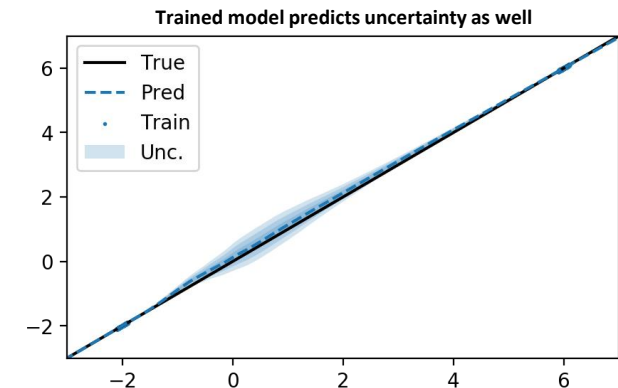
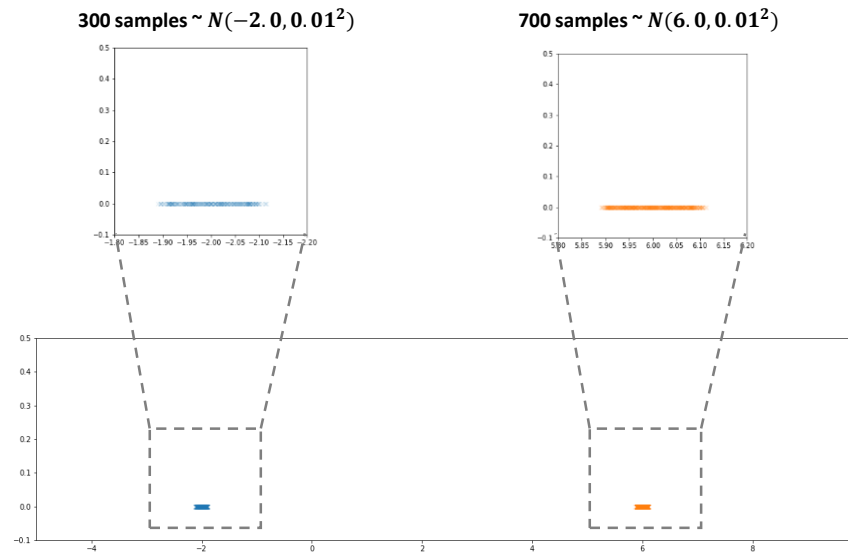
Method

- During the ambiguity set construction
 - Instead of using the fixed distance (like Wasserstein-based one),
 - Create the ambiguity set based on the uncertainty level, which can be obtained from the perception model.

Experiments

1) Train the position of the obstacles by the observed samples.

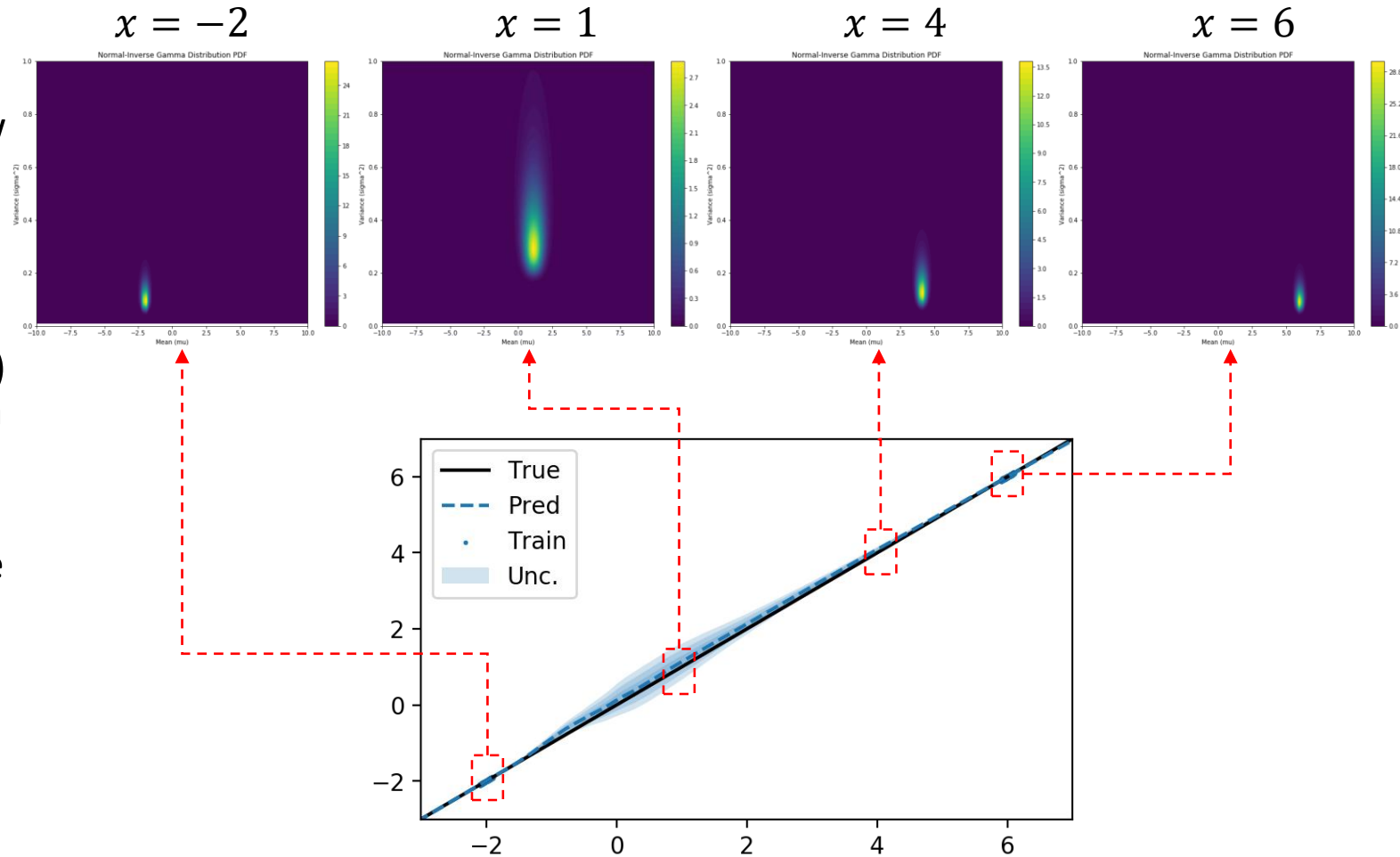
- We use the evidential deep regression to represent the amount of the model uncertainty based on the evidence of the trained samples. [7]
- Model is simple regression model, which predicts $y=x$.
- But the training samples are observed only at $x=-2$ and $x=6$.



Experiments

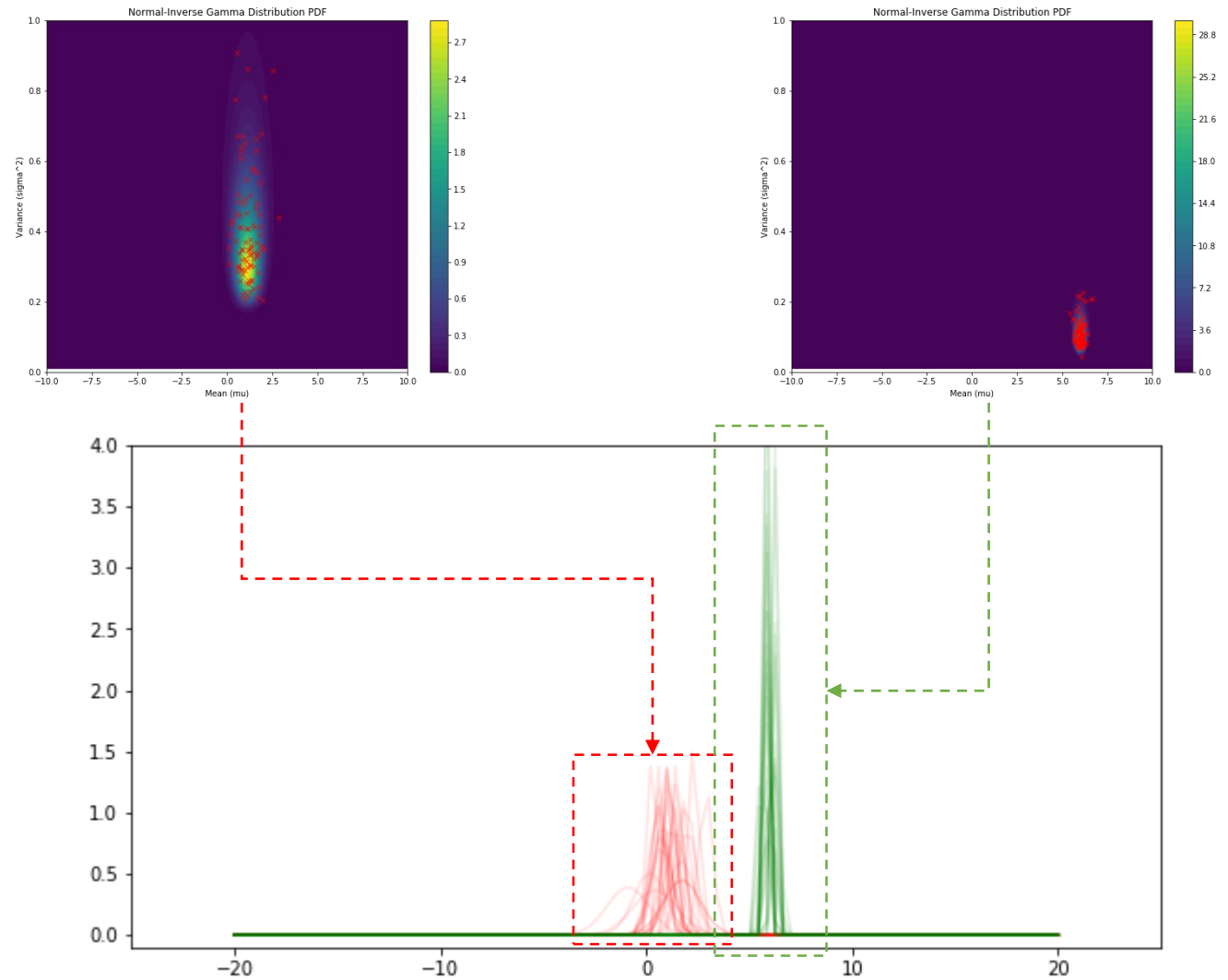
2) Prediction results

- Certain samples ($x=-2$, $x=6$) has low uncertainty, but uncertain samples ($x=1$, $x=4$) has higher uncertainty.
- Their moments are widely distributed when there is little evidence (more uncertain, like $x=1$) because there was no training data for the $x=1$.
- Upper graph represents the probability density functions of the predicted ambiguity set (Normal-Inverse Gamma distribution).
 - X-axis: mean (μ)
 - Y-axis: variance (σ^2)



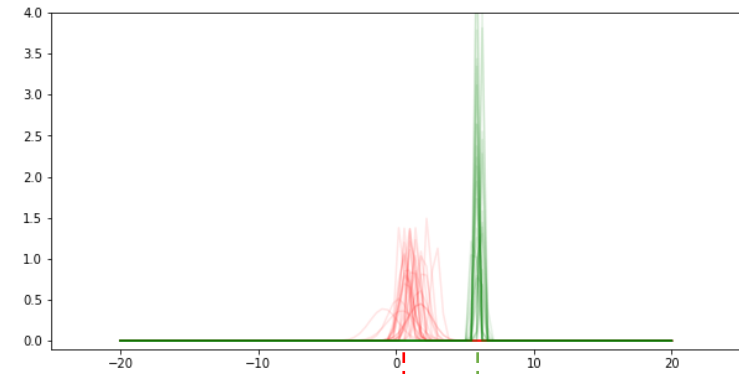
Experiments

- Based on the trained model, we can sample the distribution of the obstacle position.
- Uncertain sample ($x=1$) has large distribution of the moments μ and σ , so the variance of the distribution is larger than the certain sample's one ($x=6$).



Experiments

- Comparison with the previous work
- Compared to the previous work which uses Wasserstein distance with fixed radius, we can make adaptable range of the ambiguity set based on the knowledge acquired during the model learning.

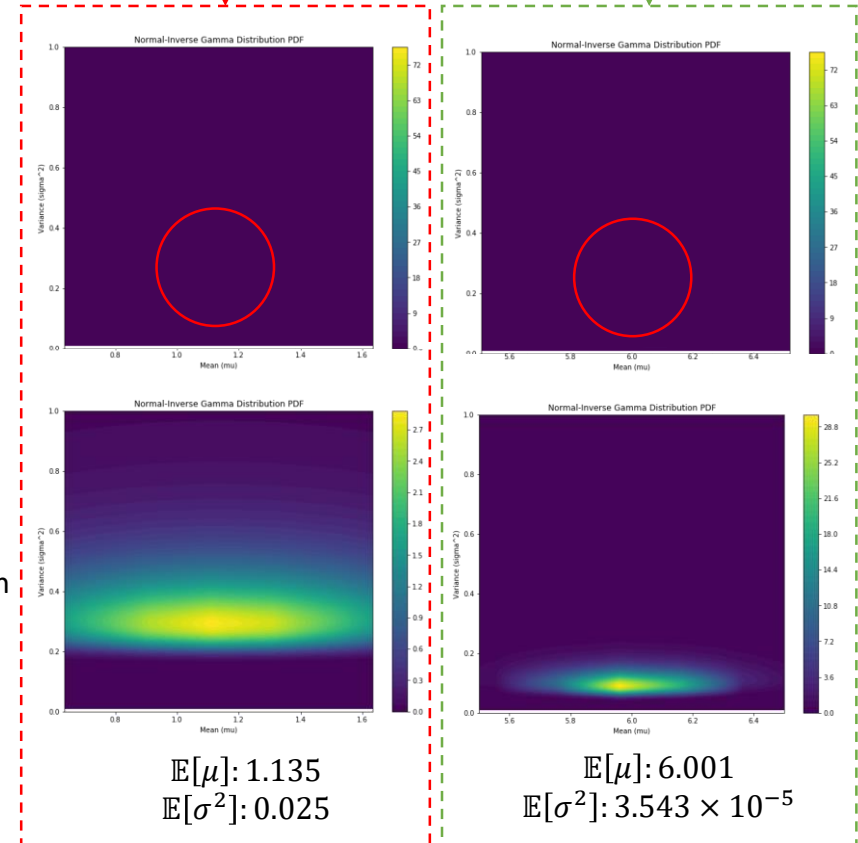


$X=1$, which is unknown during training

$X=6$, which is known during training

Wasserstein distance set
- Fixed radius

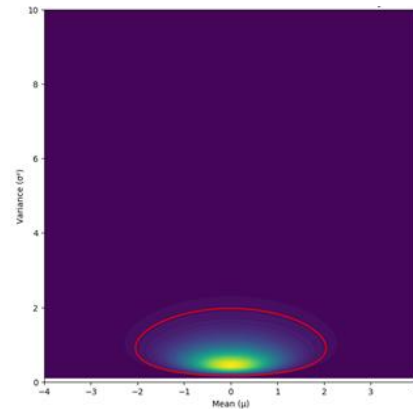
NIG set
- Adaptable range based on the trained knowledge



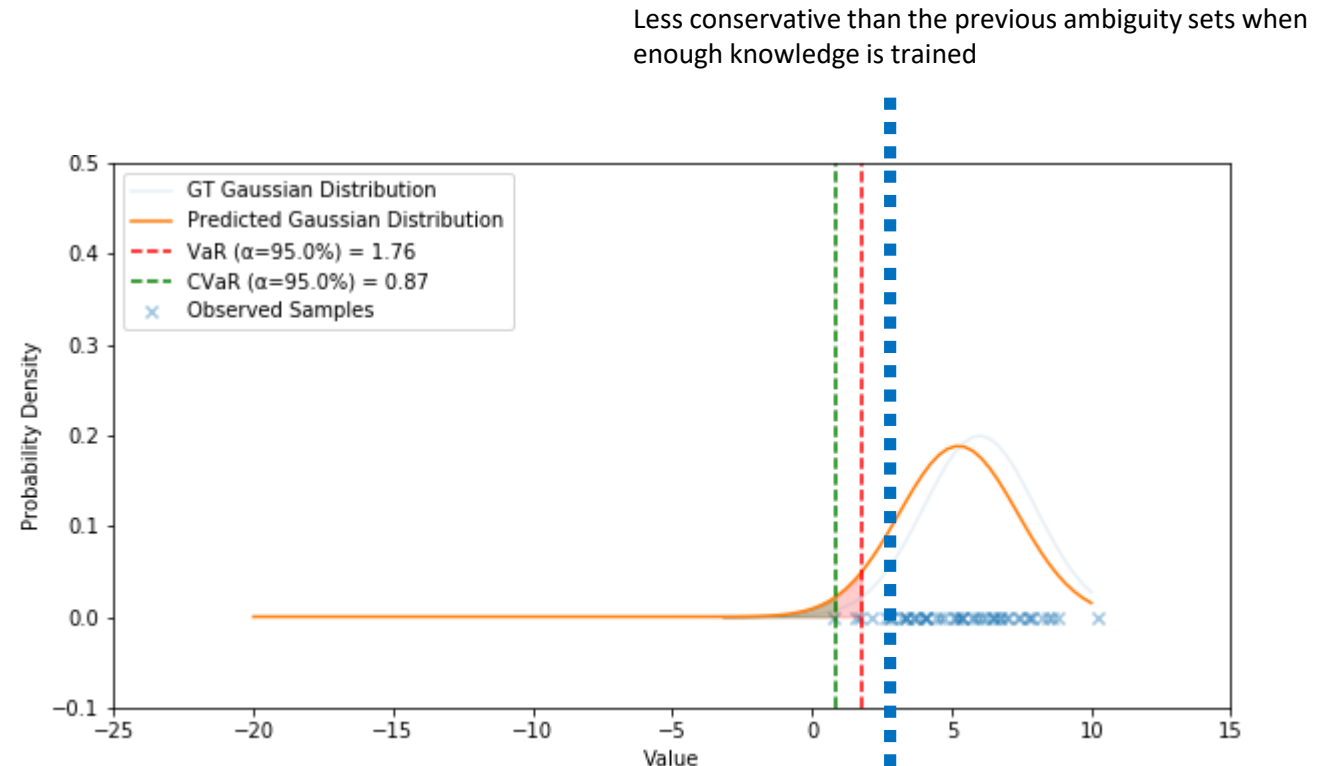
Experiments

- Next Experiment Plan

- Make a tractable solution for the DRO with our method.
- Expected results: less conservative for the well-known samples than the fixed radius method,



$\hat{x} = \epsilon\%$ cumulative probability boundary



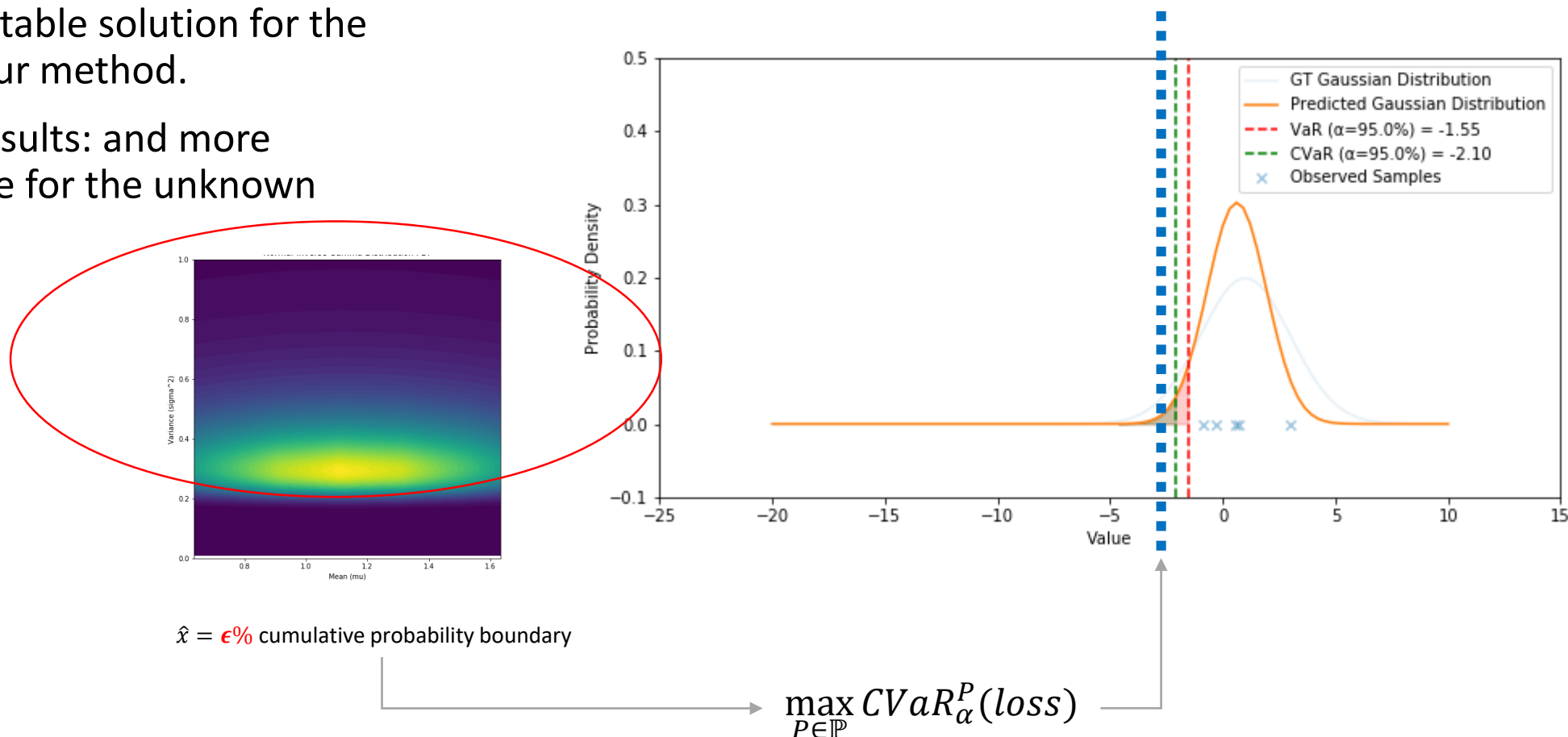
$$\max_{P \in \mathbb{P}} CVaR_{\alpha}^P(loss)$$

Experiments

- Next Experiment Plan

- Make a tractable solution for the DRO with our method.
- Expected results: and more conservative for the unknown samples.

And more conservative when little knowledge is acquired during training (uncertain)



Conclusion

- (Expected Contributions)
 - 1) Design the Knowledge-aware Distributionally Robust Optimization, which is less conservative for the well-known samples.
 - 2) Apply to the model-predictive control for the autonomous driving task.
 - 3) Reduce collision failure risk.

Current Work

- Applied KAIST EE
 - Status: 면접 준비중

- Timeline
 - ~~7. 10. (수) 원서접수~~
 - ~~7. 12. (금) 18:00 서류 제출~~
 - 8. 8. (목) 14:00 1차 합격자 발표
 - 8. 13. (화) 면접
 - 9. 26. (목) 14:00 최종 합격자 발표